Inorganic Cumulative Examination

October 24, 2008

General Chemistry can be viewed as a survey course, particularly for inorganic and physical chemistry. As such they serve as a good basis for advanced studies and students should maintain a good understanding of the topics. The following are a few questions often addressed in such textbooks.

1. Give the formula for the following:
   a. Sodium bromated
   b. Potassium dichromate
   c. Lithium nitride
   d. Calcium hypochlorite
   e. Lithium nitrite
2. For each of the following cations provide the formula of a salt that is soluble in water and one that is not. Each anion can only be used once.
   a. Barium
   b. Lead
   c. Magnesium
   d. Aluminum
   e. Silver
3. Give the geometry of the following molecules and the corresponding point group.
   a. SF_4
   b. IF_3
   c. N_2O
   d. Azide anion
   e. Thiosulfate anion
4. Define, by way of a thermodynamic cycle, lattice energy.
5. Draw a general phase diagram (P vs T) for water. Indicate what is unique about water that is evident in the diagram.
6. "van't Hoff" factors are often less than the integer values expected for strong electrolytes. Explain.
7. Aluminum was considered a precious metal in the late 1800's, two 23 year-old graduate students working independently changed that in 1886. Describe the chemistry that resulted in a collapse in the price of aluminum and the formation of Alcoa. Inc.
8. Give the elemental components of the "high temperature" superconductor most often discussed in general chemistry textbooks and readily prepared in an undergraduate laboratory.
9. Indicate which of the following ions could have high-spin and low spin configurations in an octahedral field:
   a. Mn^{4+}
   b. Mn^{3+}
   c. Co^{3+}
   d. Cu^{2+}
   e. Cr^{3+}
10. The porphyrin is a popular ligand in biochemistry and inorganic chemistry. Draw it.